INCH-POUND
MIL-DTL-32594
2 January 2018
SUPERSEDING
O-S-642G
7 May 1993
(See 6.6)

DETAIL SPECIFICATION

# SODIUM PHOSPHATE, TRIBASIC; ANHYDROUS, DODECAHYDRATE, AND MONOHYDRATE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers technical grade anhydrous, dodecahydrate, and monohydrate tribasic sodium phosphate intended for boiler water treatment and various washing, cleaning, and scouring processes (see 6.1).

1.2 Classification. The tribasic sodium phosphate covered by this specification is of the following types, as specified (see 6.2):

a. Type I - Anhydrous ($Na_3PO_4$).

b. Type II - Dodecahydrate ($Na_3PO_4 \cdot 12H_2O$).

c. Type III - Monohydrate ($Na_3PO_4 \cdot H_2O$).

1.3 Part or identifying number (PIN). PINs to be used for the tribasic sodium phosphate acquired to this specification are created as follows:

| **M** | **32594** | **-** | **X** | **-** | **XXX** |
|---|---|---|---|---|---|
| Prefix for Military Specification | Specification Number | | Type (see 1.2) | | Unit of Issue in Pounds |
| Examples: M32594-I-001, M32594-II-025 | | | | | |

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-536 - Soap and Soap-Products (Including Synthetic Detergents), Sampling and Testing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331 - Zero Acceptance Number Sampling Plans

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D501 - Standard Test Methods of Sampling and Chemical Analysis of Alkaline Detergents

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The material, as received, shall conform to the chemical and physical characteristics of table I and the particle size distribution of table II when tested as specified in 4.1.

TABLE I. Chemical and physical characteristics.

| Characteristic | Type I (percent) | Type II (percent) | Type III (percent) |
|---|---|---|---|
| Sodium phosphate, tribasic, anhydrous ($Na_3PO_4$) (minimum) | 97.0 | ---- | ---- |
| Sodium phosphate, tribasic, dodecahydrate ($Na_3PO_4 \cdot 12H_2O$) (minimum) | ---- | 97.0 | ---- |
| Sodium phosphate, tribasic, monohydrate ($Na_3PO_4 \cdot H_2O$) (minimum) | ---- | ---- | 97.0 |
| Total alkalinity to methyl orange, as sodium oxide ($Na_2O$) (range) | 36.0 - 40.0 | 16.0 - 19.0 | 32.0 - 37.0 |
| Phosphates as phosphoric anhydride ($P_2O_5$) (minimum) | 39.5 | 18.1 | 36.0 |
| Matter insoluble in water (maximum) | 0.1 | 0.1 | 0.1 |
| NOTE:<br>1. "----" indicates that the test does not apply. | | | |



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TABLE II. Particle size distribution.

| Matter retained on | Type I (percent) | Type II (percent) | Type III (percent) |
|---|---|---|---|
| U.S. No. 20 sieve (powder) | 1.0 maximum | ---- | None |
| U.S. No. 100 sieve (powder) | 10.0 maximum | ---- | 10.0 maximum |
| U.S. No. 10 sieve (granular) | 5.0 maximum | ---- | 5.0 maximum |
| U.S. No. 100 sieve (granular) | 80.0 minimum | ---- | 40.0 minimum |
| U.S. No. 10 sieve (all forms) | ---- | None | ---- |
| U.S. No. 100 sieve (all forms) | ---- | 90.0 minimum | ---- |

NOTES:
1. "----" indicates that the test does not apply.
2. "None" indicates that no matter shall be retained on the designated sieve.

3.2 Forms of types.

3.2.1 Type I. The anhydrous tribasic sodium phosphate shall be powder or granular in form, as specified (see 6.2).

3.2.2 Type II. The dodecahydrate tribasic sodium phosphate shall be granular, flake, or crystalline in form, as specified (see 6.2).

3.2.3 Type III. The monohydrate tribasic sodium phosphate shall be powder or granular in form, as specified (see 6.2).

3.3 Appearance. The finished product shall be white, uniform, free flowing, and free from foreign matter.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall be performed in accordance with inspection provisions set forth herein. The characteristics set forth in table I and table II, when tested in accordance with table III, shall constitute the minimum inspections to be performed. For purposes of sampling, an inspection lot shall consist of all material submitted for inspection and delivery at one time. Unless otherwise specified (see 6.2), sampling shall be in accordance with MIL-STD-1916, ASQ H1331, or a comparable zero-based sampling plan. Failure of any test, by any sample, shall be cause for rejection of the lot represented.

TABLE III. Verification of the end item.

| Characteristic | Requirement | Test/inspection method | Results recorded as |
|---|---|---|---|
| Sodium phosphate, tribasic | Table I | ASTM D501 | Average of two determinations to nearest 0.1% |
| Total alkalinity to methyl orange, as $Na_2O$ | Table I | ASTM D501 | Average of two determinations to nearest 0.1% |
| Phosphates as $P_2O_5$ | Table I | ASTM D501 | Average of two determinations to nearest 0.1% |
| Matter insoluble in water | Table I | ASTM D501 | Average of two determinations to nearest 0.1% |
| Sieve tests | Table II | FED-STD-536, method 2101 | Average of two determinations to nearest 0.1% |
| Appearance | 3.3 | Visual | ---- |



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Tribasic sodium phosphate covered by this specification is intended for boiler water treatment in addition to washing, cleaning, and scouring with or without soap when a fairly strong alkaline material is required. Protection of the eyes and skin is recommended. Items acquired under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2 and 3.2).

c. The specific issue of individual documents referenced (see 2.2.1 and 2.3).

d. Form of tribasic sodium phosphate required (see 3.2.1, 3.2.2, and 3.2.3).

e. Sampling plan in accordance with MIL-STD-1916 at verification level IV or ASQ H1331 at an AQL of 1.0 (see 4.1).

f. Packaging required (see 5.1).

g. Safety Data Sheets (SDSs), when required (see 6.5).

6.3 National stock numbers (NSNs). Table IV provides a list of NSNs assigned which correspond to this specification. This list may not include all possible NSNs associated with this document.

TABLE IV. List of applicable NSNs.

| NSN | Type | Unit of issue | Nomenclature | Part number |
|---|---|---|---|---|
| 6810-00-141-6078 | Type I | 1-pound bottle | Sodium Phosphate Tribasic, Anhydrous | M32594-I-001 |
| 6810-00-141-6080 | Type I | 100-pound drum | Sodium Phosphate Tribasic, Anhydrous | M32594-I-100 |
| 6810-01-038-1175 | Type II | 100-pound drum | Sodium Phosphate Tribasic, Dodecahydrate | M32594-II-100 |
| 6810-01-082-5415 | Type II | 25-pound drum | Sodium Phosphate Tribasic, Dodecahydrate | M32594-II-025 |

6.4 International standardization agreement implementation. This specification implements ABC-STD-6, Boiler Water Treatment Chemicals. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.



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6.5 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.6 Use of existing stocks. This specification supersedes O-S-642. Trisodium phosphate chemicals made to O-S-642 may continue to be used for their respective purposes until stocks are exhausted. New acquisitions will be made in accordance with this specification.

6.7 Subject term (key word) listing.

Cleaner

Water treatment

## CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH
Air Force – 68
DLA – GS

Review activities:
Army – AR, AV, EA, GL, MD, MR, TE
Navy – AS, MC
Air Force – 03, 11, 20
DLA – GS8

Civil agency
GSA – FAS

Preparing activity:
Navy – SH
(Project 6810-2017-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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